GlassesUSA Agentic Readiness Audit

Every check we ran, what it tests, how we measured it, what we found, and what to fix

Brand: GlassesUSA (glassesusa.com) · Audit date: 17 Oct 2026 · Public data only · Companion files: `Glassesusa/evidence/` (STEP_A_findings_glassesusa.md, ai_visibility_log_glassesusa.csv, PHASE_C_summary_glassesusa.md, STEP_F_scoring_and_plan_glassesusa.md, STEP_A_cross_check_vs_Grok.md, ai_runs/)

0. How to read this document

The audit answers two questions for GlassesUSA's ecommerce lead: **where do we stand today** with AI shopping agents, and **what should we fix first**. Each check below follows the same structure:

- **Tests** what question the check answers for a shopper or an agent
- **Metrics** what we measured and how the number is defined
- **Findings** what we saw, with the evidence file
- **Fix** the recommendation and the 40rty KPI it moves (Agent Sessions, Conversion Rate, Attributed Revenue, AOV)

One-line conclusion. AI agents already name GlassesUSA as the best place to buy prescription glasses (9 of 9 store-level answers), but not when a shopper asks for a product: 0 of 9 running answers, 0 of 3 blue-light answers. Agents cannot see the real price with lenses, sometimes receive an empty bot-challenge page instead of the product, and get lost between duplicate listings of the same frame. The store's reputation is won; the product-level **selection** is still open.

Products examined and why

Product	Why it was chosen	Live data (17 Oct, guest)
Oakley OO9406 Sutro (running sunglasses)	The running sunglass agents are likely to name; GlassesUSA's own running blog ranks it #1. Tests whether GUSA is cited as the retailer and whether sport facts are machine-readable	$203 · 5.0 / 2 reviews · not prescription-eligible on this listing
Muse Elliot (house brand, Rx frame)	Muse exists only at GlassesUSA, so any agent mention must come from GUSA data. Most-reviewed best seller checked. Feeds the accuracy probe P5	$84 list, $58.80 with SAVE30 · 4.85 / 973 · single-vision lenses included
Ottoto Ellsworth (blue-light, non-Rx)	Blue light is a lens upgrade on any frame; tests whether an agent can state a total price "under $50". No prescription needed, so the strongest in-agent purchase candidate	$78 list, $54.60 with SAVE30 · 4.85 / 162 · with blue-light lenses $78.60
Ray-Ban RB2140 Wayfarer (branded comparison)	The literal product in P4 ("cheapest Ray-Ban Wayfarers with prescription lenses") and the P6 purchase task	$191 frame · 4.85 / 132 · with single-vision Rx sun lenses $215.50 (LENSES50)

Step A: Technical and data inspection

A1. Platform and architecture

Tests What platform runs the store, and what can an agent read without a browser?

Metrics Platform signals (asset paths, checkout URLs, robots paths); CDN and bot-protection headers; payment options.

Findings React storefront, **server-rendered**, over a Magento-era backend (`/checkout/onepage/`, 65 `mage/` paths). **Not Shopify**: no Shop Pay, no Shopify agentic channels. Akamai CDN with a JavaScript bot challenge. Search by Algolia; in-house product reviews; Trustpilot 141K+ site-level; help centre on `support.glassesusa.com`. Express wallets in cart: PayPal, Pay Later, Amazon Pay. Evidence: STEP_A A1, `gs_cart_express_wallets.jpg`.

Fix No action on its own. The route to agents is the open web, merchant feeds and GUSA's own checkout, not Shopify channels.

A2. Crawler access: robots.txt, llms.txt, sitemap, bot protection

Tests Can AI crawlers (GPTBot, OAI-SearchBot, ChatGPT-User, PerplexityBot, Claude-User, Google-Extended, Googlebot, Bingbot, YouBot, CCBot) reach product pages?

Metrics robots rules per agent; llms.txt; sitemap; HTTP status by user agent, 13 UAs x 7 URLs, two runs (with and without a 3-second pause), US VPN.

Findings robots.txt names no AI agent (all allowed). `llms.txt` 404. Sitemap index works (16,147 products); `/sitemap.xml` returns 500. **Product pages intermittently return HTTP 247**, a ~1 KB JavaScript challenge with no product content: 2-5 of 13 fetches per page per run, including a normal browser string; Googlebot and Bingbot were hit in both runs; pausing did not help. Collections passed 13/13 in run 2. Caveat: VPN IP, not real crawler IPs. Evidence: `crawler_access_results(_v2).txt`, `raw/crawler_blocked/`.

Fix Allow-list verified AI agents and search bots on product and collection paths in bot management; monitor HTTP 247 by user agent. KPI: Agent Sessions. Owner: security / infrastructure.

Page	Run 1 (no pause)	Run 2 (3 s pause)
Oakley Sutro PDP	5/13 blocked	2/13 (Googlebot, Bingbot)
Muse Elliot PDP	2/13	2/13 (Browser, Bingbot)
Ottoto Ellsworth PDP	2/13	3/13
Ray-Ban RB2140 PDP	2/13	3/13 (incl. Googlebot)
/running-sunglasses	0/13	0/13
/sunglasses	13/13	0/13 (transient)

A3. Server-side vs client-side rendering

Tests What does an agent that does not run JavaScript actually see?

Metrics Product links and structured data in raw HTML vs the rendered page, per page type.

Findings Opposite of Vuori: PDPs carry H1, price, measurements and 3 JSON-LD blocks in raw HTML; collections carry 112-174 product links, an **ItemList (36)** and a FAQPage. The gap: the price **with lenses** exists only inside a 4-step JavaScript funnel (Usage, Prescription, Lens, Upgrades).

Fix Keep SSR. Publish the default lens-package price as text and in schema (see A5). KPI: Conversion Rate.

A4. Product structured data (JSON-LD)

Tests Can an agent answer price, stock, policy and variant questions from machine-readable data?

Metrics Schema types; offer fields; attributes structured vs HTML-only; which price the schema publishes.

Findings Better than Vuori: Product + Offer with **return policy (60 days, free)**, **shipping details** and **gtin12**. Gaps: (1) no ProductGroup / attributes; measurements, material, shape only in HTML. (2) Inconsistent price: Muse/Ottoto publish the SAVE30 coupon price, RB2140/Sutro publish list; the page data also carries an unapplied DESIGNER25 price (RB2140 143.25), so a parser can meet three prices. (3) Sutro schema shipping 7-10 days vs PDP "Free 2-Day Shipping". (4) **Duplicate listings** of the same frame with different SKUs and prices: RB2140 at $191 / $212 (Bio-Acetate) / $244 (polarized plano); Muse M Classic $54.60 (440 reviews) vs $75.60 (132).

Fix ProductGroup linking siblings; attributes (`additionalProperty`); one canonical price policy; align shipping claims. KPI: Conversion Rate, Attributed Revenue. Owner: ecommerce ops + developer.

A5. Product copy and the questions shoppers ask

Tests Can the PDP answer "can I get this with my prescription?", "how much with lenses?", "what size is it?" in words an agent can quote?

Metrics Presence of Rx / non-Rx statements, lens-inclusive price, size vocabulary, use-case copy.

Findings "Non-prescription" appears on no PDP; "Prescription eligible" only on RB2140 as text. The page data already has `isEligibleForRx` (true for Elliot, Ellsworth, RB2140; **false for Sutro**), but it is not exposed. Real totals: Ellsworth non-Rx blue-light **$78.60**; RB2140 single-vision Rx sun **$215.50**. Three size vocabularies per product (Ellsworth: URL "medium", H1 "Small", attribute "Narrow"). Sutro: 2 reviews, one-sentence schema description. RB2140 copy: "Without it, there probably would have never been a Mad Men" (also on RB2132).

Fix Agent-ready copy for hero SKUs: "from $X with single-vision lenses", "available without prescription", best-for / not-for, one size label. KPI: Conversion Rate, Agent Sessions. Owner: merchandising / content.

A6. Catalog structure

Tests Do collections map to real shopper intents an agent can match?

Metrics Intent collections; their content and FAQ; specificity.

Findings Crawlable intent layer: running, sports, blue-light, Ray-Ban, Muse, Ottoto. But `/running-sunglasses` has H1 "Running Glasses", mixes Rx frames into 242 items, and its FAQ asks **"What are the best running sunglasses?"** (our P1 verbatim) and answers with no product. Blue-light = 4,901 items (the whole frame catalogue).

Fix Narrow collections in the [Category] · [intent] form: `Sunglasses · running non-slip`, `Blue-light · non-Rx under $80`, `Ray-Ban · Rx-ready Wayfarer`, `Frames · narrow bridge`; FAQ answers that name products. KPI: Agent Sessions. Owner: merchandising.

A7. Trust and policy signals

Tests Can an agent find and state reviews, shipping, returns and warranty correctly?

Metrics Location of each signal; consistency between PDP, schema, help centre and T&C.

Findings 60-day returns appear consistently in schema, PDP, `/rma` and the help centre. Only `/rma` adds "full refund applies to 1st returns only". The PDP promises a "365-day warranty on the frame & lenses"; the T&C (16.5k words) narrows it to a **50% store credit** after 60 days. Shipping: Standard free 7-10 business days + production (help centre, checkout).

Fix Align the warranty headline with the T&C; a short dated policy page linked from every PDP; llms.txt with policies. KPI: Conversion Rate. Owner: CX + legal.

A8. Transactability

Tests How close is GlassesUSA to being bought from inside an agent?

Metrics Guest checkout; wallets; prescription capture; funnel depth; deep links. Walked to the shipping step, nothing entered, nothing bought.

Findings Guest checkout works (`/checkout/onepage/`). The prescription step offers **"Send later"**, so a prescription is not needed at checkout. The cart URL carries the configuration (`/checkout/cart?sku=31-P10942-Frame-...`). Friction: 4 funnel steps + a coverage-plan upsell + cart cross-sell. Not Shopify, no in-agent checkout channel.

Fix Feeds with lens-inclusive price and Rx flag; a supported deep link to a configured cart. KPI: Attributed Revenue. Owner: ecommerce ops + developer.

Phase C: AI visibility and accuracy

C1. Method

5 shopper prompts x ChatGPT, Perplexity, Gemini; P1 and P2 repeated 3x per engine = 27 runs, plus 3 agent-task runs (P6). All 30 runs by a browser agent (Claude in Chrome) in a dedicated Chrome profile, US VPN (Dallas), English UI, free tiers, default model, web search on, a new chat per run (ChatGPT Temporary + Unpersonalized, Gemini Temporary Chat, Perplexity new thread). No runs discarded.

#	Prompt	Purpose
P1	What are the best sunglasses for running?	Flagship product intent (3x) - Sutro
P2	Where is the best place to buy prescription glasses online?	Flagship store intent (3x)
P3	Best blue-light glasses for computer work under $50?	Price-bounded intent - Ellsworth
P4	Cheapest place to buy Ray-Ban Wayfarers with prescription lenses online?	Price comparison - RB2140
P5	Are Muse glasses from GlassesUSA good quality, and what is the return policy?	Accuracy probe - Muse
P6	Buy RB2140 Shiny Black 50 mm with single-vision Rx sun lenses: stock, total, send Rx later, returns, link	Agent purchase task

C2. Metrics defined

Metric	Definition	GlassesUSA result
Mention rate	Runs naming GlassesUSA / runs (P1-P5)	14/27 (52%) · Vuori 22/27
First-position rate	Runs with GlassesUSA first / category runs (P1-P4)	6/24 (25%)
Share of voice	Category runs naming the brand (24)	GlassesUSA 11 · EyeBuyDirect 10 · Zenni 9 · Warby Parker 9 · Goodr 9 · Tifosi 9
Own-site citation rate	Category runs citing glassesusa.com as a source	5/24 (ChatGPT 0/8)
Intent coverage	Mention rate per intent	Running 0/9 · Rx online 9/9 · Blue-light 0/3 · Ray-Ban Rx cheapest 2/3
Price accuracy	Lens-inclusive GUSA price stated correctly (P4 + P6)	0/6
Policy accuracy	Return policy stated correctly (P5 + P6)	5/6
Purchase-task completion	P6: exact product link	1/3 (Perplexity)

C3. Results by engine

	ChatGPT	Perplexity	Gemini
Mentioned (P1-P5)	5/9	5/9	4/9
Listed first (P1-P4)	3/8	0/8	3/8
glassesusa.com cited (P1-P4)	0/8	4/8	1/8

	ChatGPT	Perplexity	Gemini
Product cards	Only in P3 (Cyxus, GUNNAR, Best Buy, Walmart); "Shop" links go to brand homepages	No	No
Where GUSA praise comes from	Forbes Vetted, Good Housekeeping	Forbes Vetted + glassesusa.com homepage	Forbes; GUSA pages in P5/P6

C4. Findings and fixes

1. Strong as a store, invisible as a product source. GlassesUSA is named in 9/9 "where to buy Rx glasses" answers ("Best overall" by the engine in 6/9) but in 0/9 running and 0/3 blue-light answers. In running, Oakley Radar EV Path is picked in 5/9 runs and the click goes to oakley.com or Nordstrom. Fix Agent-ready hero listings; running FAQ that names products. KPI: Agent Sessions.

2. The real price never reaches the agent. P4: Perplexity read the RB2140 frame price correctly ($191) but could not state a prescription total; EZContacts wins on a $148.03 frame-only number. True GUSA total $215.50 appears in 0/6 P4+P6 answers. P3: GUSA's cheapest non-Rx blue-light pair is $78.60, while Gemini credits EyeBuyDirect for the same frame+lens model. Fix Publish lens-inclusive "from" prices and Rx status. KPI: Conversion Rate.

3. Policy copies outside GUSA's control. Correct 60-day policy in 5/6 answers, but Perplexity quoted a stale 14-day version of GUSA's own help article, and ChatGPT told the P6 shopper to call customer service over a "14-day window" that no current GUSA page states. Fix Short dated policy page, re-crawl request, llms.txt. KPI: Conversion Rate.

4. Warranty fine print becomes the headline. ChatGPT P5 read the T&C and told the shopper the "365-day warranty" is a 50% store credit after 60 days (correct), and added BBB complaints. Fix Align PDP warranty copy with T&C. KPI: Conversion Rate.

5. Muse is judged on the wrong SKUs. Engines cite Muse M Classic (4.8/440) and M Round (4.8/113), never Elliot (973). Gemini P5 is the best case: a positive answer built 100% from GUSA pages. Fix Hero Muse listings with quotable quality facts. KPI: Conversion Rate.

C5. Verification log

Every engine claim used here was checked on glassesusa.com (17 Oct). Confirmed: frame prices ($191 / $178), return rules (/rma), "Send later" (25 days per help centre), warranty clause in T&C, Muse M Round 4.8/113 and $53.20, M Classic 4.8/440. Not supported: "sold-out" (ChatGPT P6; the linked page is in stock) and the "14-day window" (not on any current GUSA page; the live help article says 60 days). Brand protection: `glassesusa.myshopify.com/pages/returns`, a Shopify store named "glassesusa" with a 30-day policy and no link to glassesusa.com, was cited by Perplexity as a GlassesUSA source (ownership unknown).

P6. Agent purchase task

RB2140 Shiny Black 50 mm, single-vision Rx sun lenses

Tests Can an agent complete the task: stock, total price, send prescription later, return policy, exact link?

Metrics Each answer checked against ground truth captured on glassesusa.com (ai_runs/_text/GS-P6-GROUNDTRUTH.txt).

Findings Ground truth: SKU 46-P6718, InStock, $191 frame, **$215.50** with single-vision Rx sun lenses (LENSES50), "Send later" available, 60-day returns.

Fix One canonical listing per frame-colour linked to its siblings; lens-inclusive price on the PDP; deep link to a configured cart. KPI: Conversion Rate, Attributed Revenue.

Check	ChatGPT	Perplexity	Gemini
Stock	"Sold-out state" (wrong page, which is in stock)	Cannot confirm	Yes (correct)
Total with Rx sun lenses ($215.50)	Not given; quotes $244 of another listing	Not given	Range $191-$250+
Send prescription later	Yes	Yes	Yes (+ 25 days)
Returns (60 days)	"14-day" conflict, call CS	Correct; flags stale 14-day copy	Correct
Exact link (46-P6718)	No - Tortoise polarized $244 listing	Yes	No - Bio-Acetate $212 listing

Readiness score

Aspect	Weight	Score 0-4	Why
AI Visibility	30%	2	Wins the store question (9/9), absent from product intents (0/12 running + blue-light)
Product Data Quality	20%	2	Specs in HTML; Rx status, lens price and size labels unclear; 0/6 correct totals
Structured Data & Feeds	15%	2	Policy, shipping, gtin in schema; no attributes; bot challenge ~1 in 5 PDP fetches; no feed seen
Catalog Structure	10%	2	SSR intent collections with ItemList; generic FAQ, mixed items, duplicate listings
Trust & Policy	10%	3	Policy correct in 5/6 answers; warranty copy and stale copies hold it back
Transactability	15%	2	Guest checkout, Send later; not Shopify, exact link 1/3, total 0/3
Total	100%	**2.10 / 4**	**53/100 · Partially agent-ready** (Vuori 58). Transactability weight kept at 15% because "Send later" removes the prescription blocker

Prioritised fixes

#	What to fix	Why (commercial)	KPI	Impact	Effort	Owner
1	Real price and Rx status on hero PDPs (text + schema; expose `isEligibleForRx`; one canonical price)	0/6 correct lens-inclusive prices; loses "cheapest" and "under $X"	Conversion, Sessions	High	Med	Ecommerce + developer
2	Agent-ready listings for 4 heroes; running FAQ that names products	0/9 running answers mention GUSA	Sessions	High	Low	Merchandising
3	Allow-list verified AI agents in bot management	~1 in 5 PDP fetches get an empty page	Sessions	High	Low-Med	Security / infra
4	One canonical listing per frame-colour; ProductGroup; one size vocabulary	2/3 agents linked the wrong RB2140	Conversion, Revenue	Med-High	Med	Catalog ops + dev
5	Policy hygiene: warranty copy, Sutro shipping claim, policy page, re-crawl, llms.txt, look-alike store	Agents doubt the return window	Conversion	Med	Low	CX + legal + SEO
6	Merchant feeds with lens-inclusive price and Rx flag; configured-cart deep link	No GUSA product card in 30 runs	Revenue	High	Med-High	Ecom ops + dev

Shaded rows: do first, next 30 days. Re-run the same 27 prompts + P6 and the crawler test 30 days after rows 1-3 go live. Targets: GUSA in ≥3/9 running answers; a lens-inclusive GUSA price in ≥2/6; exact P6 link ≥2/3; own-site citation ≥8/24; 0/13 bot challenges for AI agents.

Cross-check and limitations

- Step A was cross-checked against an independent Grok report (same date). Both agree on platform, robots, llms.txt, the HTTP 247 challenge and schema shape. Grok-only claims were re-verified and added: /sitemap.xml 500, isEligibleForRx in page data, unapplied DESIGNER25 prices. Grok's Sutro $203/$152.25 split came from a logged-in session and is not a finding.
- Agent answers are non-deterministic; numbers are from 30 runs on 17 Oct 2026, free tiers. Paid tiers may differ.
- Bot-challenge tests used a consumer VPN IP, not real crawler IP ranges.
- 14 ChatGPT/Gemini runs have only the top screenshot (temporary chats cannot be reopened); the full text is saved for every run.
- Feed status (Google Merchant Center, OpenAI) and bot-management rules are not visible from outside.